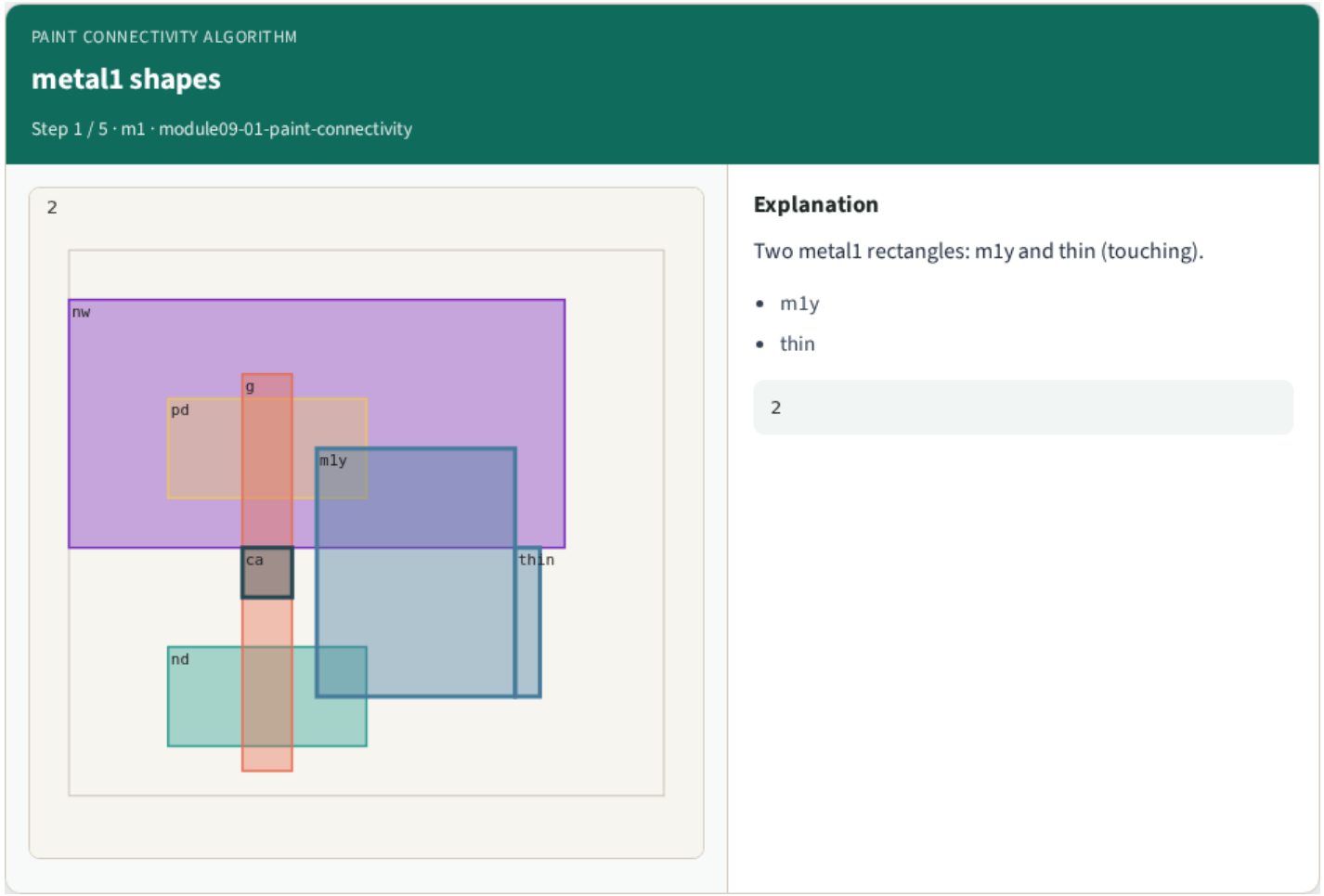
Why algorithms

Flood-fill metal1 nets; contact bridges poly and metal1

The idea

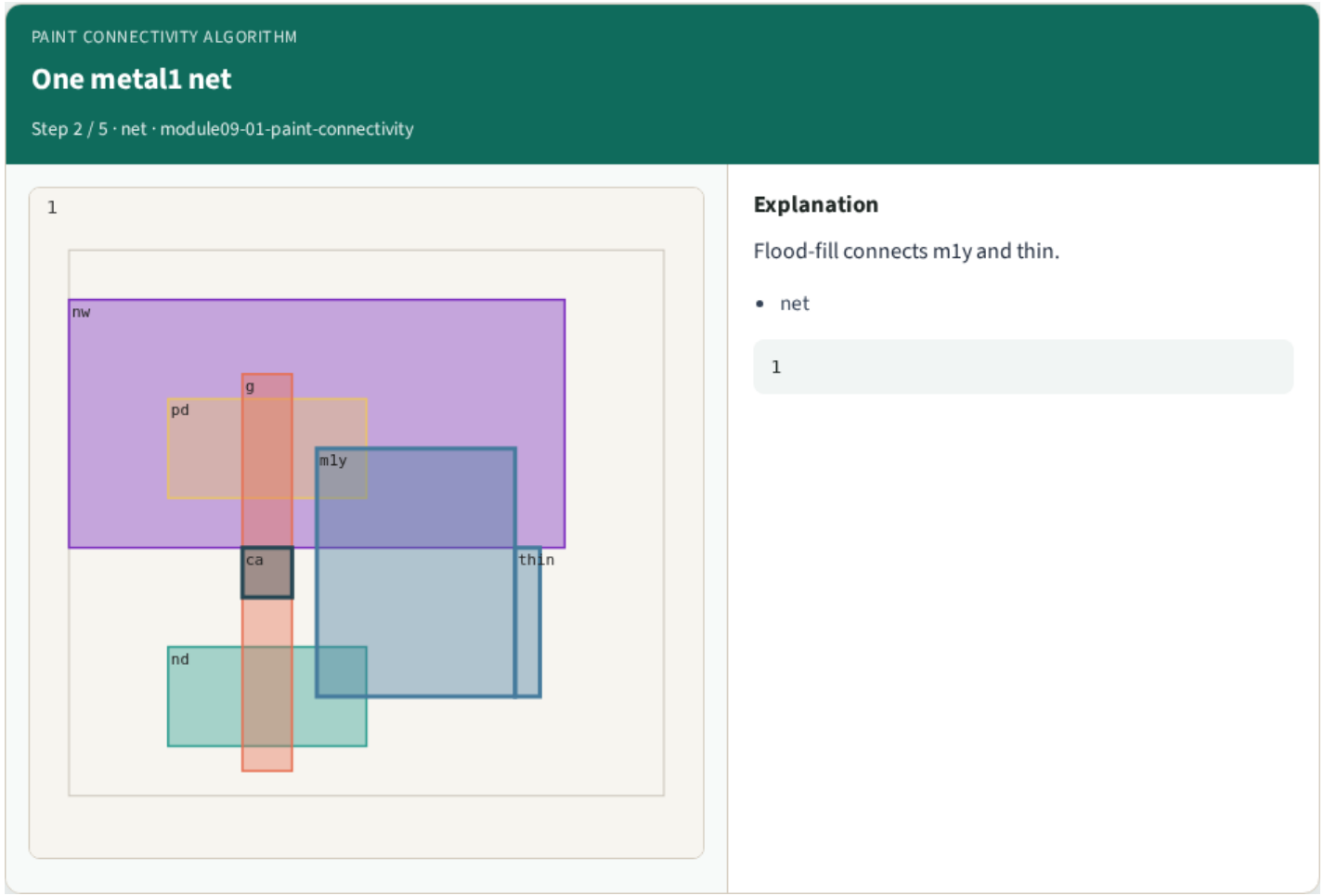
- GOLDEN: metal1_net_shapes=2; contact_count=1

metal1 shapes



metal1 shapes

One metal1 net



One metal1 net

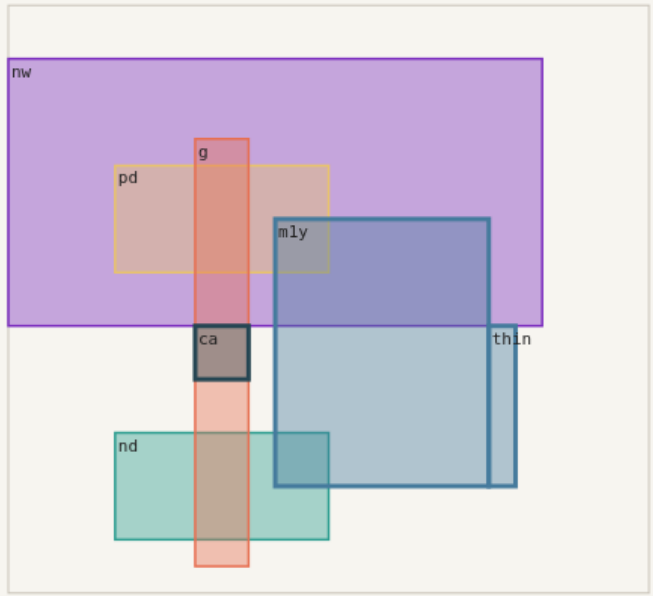
Contact ca

PAINT CONNECTIVITY ALGORITHM

Contact ca

Step 3 / 5 · ca · module09-01-paint-connectivity

1



Explanation

Contact bridges poly gate g and metal1.

- ca

```
1
```

Contact ca

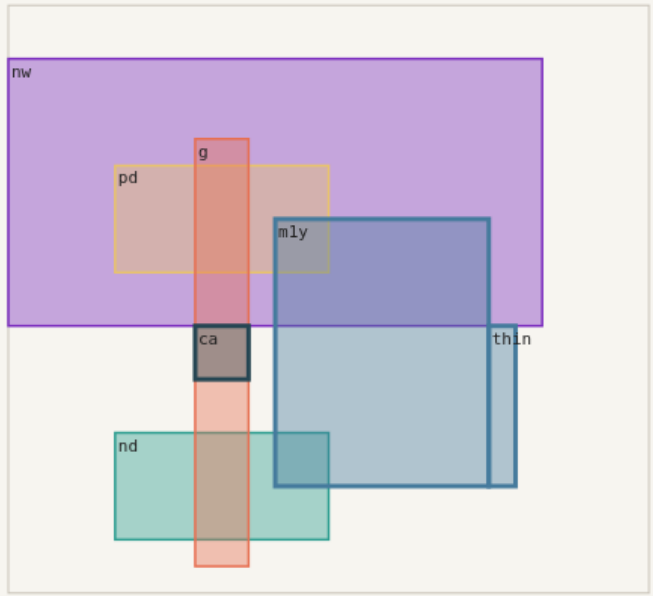
poly link

PAINT CONNECTIVITY ALGORITHM

poly link

Step 4 / 5 · poly · module09-01-paint-connectivity

poly



Explanation

poly layer participates via contact.

- g

```
poly
```

poly link

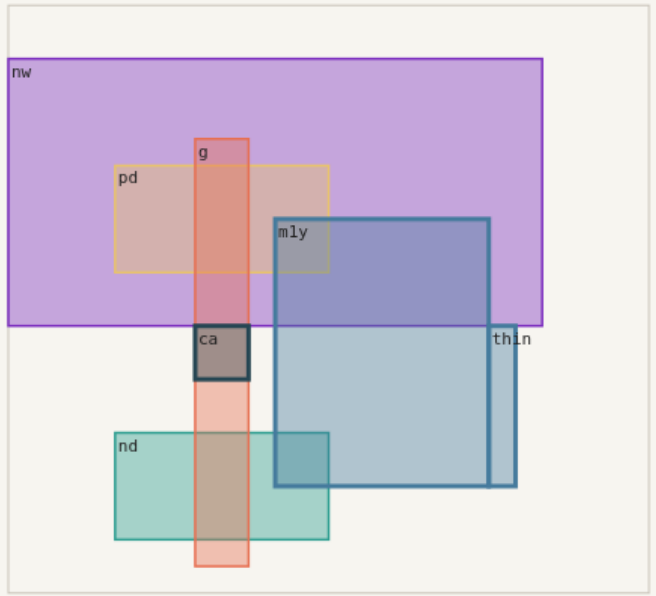
Golden

PAINT CONNECTIVITY ALGORITHM

Golden

Step 5 / 5 · golden · module09-01-paint-connectivity

2/1



Explanation

GOLDEN: metal1_net_shapes=2 contact=1.

- GOLDEN

```
2/1
```

Golden

Browser lab track

- Open **paint-connectivity**
- Run connectivity / DRC / extract / bind, then Check

Implement track

- From common/, run algo_solvers.py and partc_bind.py

Pitfalls

- Changing tiny_algo_magic without updating Part B goldens
- Ignoring the bind side

Try these

- From the common folder, run the Part C self-tests

Try these — try these

```
cd courses/learn_magic/common
python3 algo_solvers.py
python3 partc_bind.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
metal1_net_shapes=2; contact_count=1.
ok if goldens match and bind
```

Next

- Complete the quiz for this part
- Continue Part C, then internals literacy